Reg. No. | | | | | | | | | |

Question Paper Code : 12804

B.E / B.TECH DEGREE EXAMINATION, MAY / JUNE 2017

Third Semester

B.E – Computer Science and Engineering

15CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

1. What is order of growth of an algorithm?
2. Define big Oh notation.
3. Compare singly linked list and doubly linked list.
4. Suggest two applications for Stack ADT.
5. Which traversal technique of a tree is used to display the elements in sorted order?
6. Compare binary tree and threaded binary tree.
7. Write the effect of Path compression.
8. Why the hash table size is chosen as a prime number?
9. List the ways to represent the Graph.
10. Write the properties of a Minimum Spanning Tree.

PART B – (5 × 16 = 80 Marks)

11. a) i. Consider the following algorithm.

```
ALGORITHM Secret(A[0..n − 1])
//Input: An array A[0..n − 1] of n real numbers
minval← A[0]; maxval← A[0]
for i ← 1 to n − 1 do
if A[i] <minval
      minval ← A[i]
if A[i] >maxval
      maxval ← A[i]
return maxval–minval
```

What does this algorithm compute? What is its basic operation? Compute the time and space complexity. (10 Marks)

11. a) ii. Show an example for the best case, worst case and average case inputs for sequential search problem. (6 Marks)

OR

11. b) i. Design a recursive and non recursive algorithms for finding the factorial of a number. Analyze the efficiency of both the algorithms. (10 Marks)

11. b) ii. One algorithm has the computing time O(logn) and another algorithm has the computing time as $O(n^2)$. Detect which algorithm is faster. Justify your answer. (6 Marks)

12. a) i. Design a List ADT for storing sent MESSAGES in a mobile. The operations are Insert, Display messages based on mobile number. (8 Marks)

12. a) ii. Consider two lists L1 and L2. Write a routine to print the common elements present in the list L1 and L2. (8 Marks)

OR

12. b) i. Design a suitable ADT to store the ITEM NUMBER to be inspected in first come first served basis by the inspector. The design should include the basic operations. (8 Marks)

12. b) ii. Design and implement an operation to reverse a string using Stack ADT. (8 Marks)

13. a) i. Using Binary search tree construct a Supplier Contact information with Phone Number and perform the following :

- Insert a new number
- Print all numbers (10 Marks)

13. a) ii. Illustrate the construction of the expression tree using the following expression (a+b)*(c+d). (6 Marks)

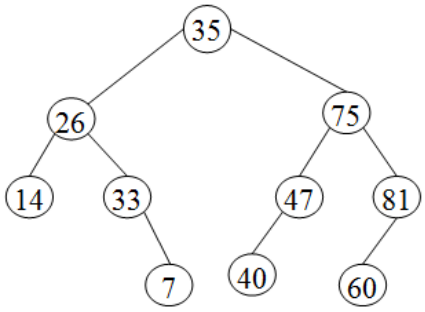
OR

13. b) i. Built a min heap with the following elements given.

45, 6, 11, 23, 5, 7, 22, 71, 15, 10.

Then perform delete min once and show min heap. (10 Marks)

13. b) ii. Is the tree below a binary search tree or not? Justify.



(6 Marks)

14. a) i. Explain in detail about how search operation is carried out in separate chaining method with an example. (6 Marks)

14. a) ii. Suppose there are 10 sets, each containing a distinct elements namely A, B, C, D, E, F, G, H, I, J. Assume the values 1, 2, 3,…. etc. for A, B, C,… etc. respectively. Perform the following sequence of set operation using union by size.
Union (A,D) Union (E,F) Union (E,H) Union (A,B) Union (G,J) Union (A,E) Union(J,A). (10 Marks)

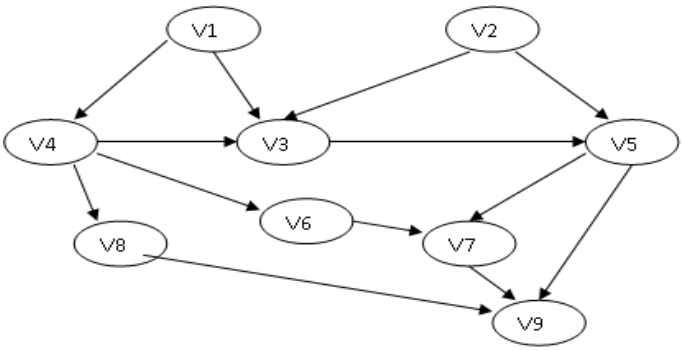
OR

14. b) i. Explain in detail about union by height operation in Disjoint set using an example. (6 Marks)

14. b) ii. Insert the following items into the hash table using quadratic probing method. Assume the table size as 13. Hash function is h(x) = x mod 10.
15, 20, 36, 7, 21, 45, 7, 22, 62, 55, 14. (10 Marks)

15. a) i. Write and explain any one Graph traversal algorithm. (8 Marks)

15. a) ii. Perform topological sort using the below graph.



(8 Marks)

OR

15. b) i. Write and explain the algorithm to find connected components in a graph. (8 Marks)

15. b) ii. Draw a graph corresponds to the following adjacency matrix representation. Show the adjacency list. Identify the edges and indegree of the vertices.

	A	B	C	D	E
A	0	1	0	1	1
B	1	0	1	1	0
C	0	1	0	1	1
D	1	1	1	0	1
E	1	0	1	1	0

(8 Marks)